In [8]:
```
#1. Read a file containing student names (one per line) and store them in a list.
with open("student.txt", "r") as file:
names = file.read().splitlines()
print("student names:", names)
```

```
student names: ['Aarav Mehta', 'Priya Shah', 'Aditya Patil', 'Riya Deshmukh', 'Neha Kulkarni', 'Omkar Joshi',
```

In [7]:
```
#2. Print names longer than 5 characters.
with open("student.txt", "r") as file:
names = file.read().splitlines()
print("name longer than 5 characters:")
for name in names:
if len(name) > 5:
print(name)
```

```
name longer than 5 characters:
Aarav Mehta
Priya Shah
Aditya Patil
Riya Deshmukh
Neha Kulkarni
Omkar Joshi
Sneha More
Karan Shah
Isha Pawar
Rahul Nair
Aditya Patil
```

In [11]:
```
#3. Count how many names start with a vowel.
with open("student.txt", "r") as file:
names = file.read().splitlines()
count = 0
for name in names:
if name and name[0].lower() in "aeiou":
count += 1
print("Number of names starting with a vowel:", count)
```

```
Number of names starting with a vowel: 5
```

In [13]:
```
#4. Save the filtered names (longer than 5 characters) into a new file.
with open("student.txt", "r") as file:
names = file.read().splitlines()
filtered_names = []
for name in names:
if len(name) > 5:
filtered_names.append(name)
with open("filtered_names.txt", "w") as file:
for name in filtered_names:
file.write(name + "\n")
print("Filtered names saved successfully")
print(filtered_names)
```

```
Filtered names saved successfully
['Aarav Mehta', 'Priya Shah', 'Aditya Patil', 'Riya Deshmukh', 'Neha Kulkarni', 'Omkar Joshi', 'Sneha More', '
```

In [16]:
```
#5. Sort the list of names alphabetically and print them in ascending order.
with open("student.txt", "r") as file:
names = file.read().splitlines()
names.sort()
print("Names in ascending order:")
for name in names:
print(name)
```

```
Names in ascending order:
Aarav Mehta
Aditya Patil
Aditya Patil
Isha Pawar
Karan Shah
Neha Kulkarni
Omkar Joshi
Priya Shah
Rahul Nair
Riya Deshmukh
Sneha More
```

In [17]:
```
#6. Print names that contain the letter "a".
with open("student.txt", "r") as file:
names = file.read().splitlines()
print("Names containing letter 'a':")
for name in names:
if "a" in name.lower():
print(name)
```

```
Names containing letter 'a':
Aarav Mehta
Priya Shah
Aditya Patil
Riya Deshmukh
Neha Kulkarni
Omkar Joshi
Sneha More
Karan Shah
Isha Pawar
Rahul Nair
Aditya Patil
```

In [20]:
```
#7. Count name frequency (including duplicates) and store results in a dictionary.
with open("student.txt", "r") as file:
names = file.read().splitlines()
frequency = {}
for name in names:
if name in frequency:
frequency[name] += 1
else:
frequency[name] = 1
print("Name frequency:")
print(frequency)
```

```
Name frequency:
{'Aarav Mehta': 1, 'Priya Shah': 1, 'Aditya Patil': 2, 'Riya Deshmukh': 1, 'Neha Kulkarni': 1, 'Omkar Joshi':
```

In [21]:

```
#8. Display each name, its length, first/last character, and palindrome status.
with open("student.txt", "r") as file:
names = file.read().splitlines()
for name in names:
if not name:
continue
print("Name:", name)
print("Length:", len(name))
print("First character:", name[0])
print("Last character:", name[-1])
print("Palindrome:", "yes" if name.lower() == name[::-1].lower() else "no")
print()
```

```
Name: Aarav Mehta
Length: 11
First character: A
Last character: a
Palindrome: no

Name: Priya Shah
Length: 10
First character: P
Last character: h
Palindrome: no

Name: Aditya Patil
Length: 12
First character: A
Last character: l
Palindrome: no

Name: Riya Deshmukh
Length: 13
First character: R
Last character: h
Palindrome: no

Name: Neha Kulkarni
Length: 13
First character: N
Last character: i
Palindrome: no

Name: Omkar Joshi
Length: 11
First character: O
Last character: i
Palindrome: no

Name: Sneha More
Length: 10
First character: S
Last character: e
Palindrome: no

Name: Karan Shah
Length: 10
First character: K
Last character: h
Palindrome: no

Name: Isha Pawar
Length: 10
First character: I
Last character: r
Palindrome: no

Name: Rahul Nair
Length: 10
First character: R
Last character: r
Palindrome: no

Name: Aditya Patil
Length: 12
First character: A
Last character: l
Palindrome: no
```

In [24]:
```
#9. Remove duplicates, sort names by length, and save them.
with open("student.txt", "r") as file:
names = file.read().splitlines()
unique_names = list(set(names))
unique_names.sort(key=len)
with open("processed_names.txt", "w") as file:
for name in unique_names:
file.write(name + "\n")
print("Duplicates removed successfully")
print("Names sorted by length:")
print(unique_names)
print("Results saved in processed_names.txt")
```

```
Duplicates removed successfully
Names sorted by length:
['Priya Shah', 'Isha Pawar', 'Karan Shah', 'Sneha More', 'Rahul Nair', 'Aarav Mehta', 'Omkar Joshi', 'Aditya P
Results saved in processed_names.txt
```

In [26]:
```
#10. Group names by their first letter and save each group in a separate file.
with open("student.txt", "r") as file:
names = file.read().splitlines()
groups = {}
for name in names:
if name:
letter = name[0].upper()
if letter not in groups:
groups[letter] = []
groups[letter].append(name)
for letter, group_names in groups.items():
filename = letter + "_name.txt"
with open(filename, "w") as file:
for name in group_names:
file.write(name + "\n")
print(filename, "created successfully")
print("All name groups saved successfully")
```

```
A_name.txt created successfully
P_name.txt created successfully
R_name.txt created successfully
N_name.txt created successfully
O_name.txt created successfully
S_name.txt created successfully
K_name.txt created successfully
I_name.txt created successfully
All name groups saved successfully
```